Abaid-ur-Rehman

Machine Learning Engineer | AI Engineer | Python Developer

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SUMMARY

Machine Learning Engineer and Python Developer with hands-on experience in AI, Machine Learning, Computer Vision, and LLM-based applications. Skilled in Python, Flask, TensorFlow, Scikit-learn, OpenCV, Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG). Passionate about building intelligent solutions and applying AI to solve real-world problems.

EDUCATION

Bachelor of Science in Artificial Intelligence 2023 – Cont.
NFC Institute of Engineering and Fertilizer Research, Faisalabad

Intermediate (Pre-Engineering), *KIPS College, Faisalabad* 2021 – 2023

Matriculation (Science), *Allied School, Faisalabad* 2019 – 2021

PROJECTS

StudyMind– AI-Powered PDF Learning Platform
Technologies: Python, Flask, SQLAlchemy, FAISS, PyMuPDF, Google Gemini/OpenAI/Anthropic APIs, Flask-Login, Flask-Mail, JavaScript, HTML/CSS
- Full-stack Flask web app that lets users chat with uploaded PDFs using RAG, with dual-answer responses (document-grounded + AI-explained), auto-generated summaries, flashcards, and quizzes. Built a pluggable multi-provider AI layer (easy switchable to any LLM via a single API key) and a full auth system.

Jarvis AI Desktop Assistant
Technologies: Python, Flask, Ollama , Speech Recognition
- Developed an AI-powered desktop assistant with voice command recognition and conversational AI capabilities.
- Implemented task automation including application control, web search, file management, and system utilities using local LLMs .

AI-Image-Generator
Technologies: Python, Flask, JavaScript, HTML5, CSS3, Pillow, Hugging Face API, JSON
- Developed a full-stack AI image generation application integrating Hugging Face models for text-to-image generation.
- Implemented prompt enhancement, image editing (rotate, flip, brightness, contrast, saturation, sharpness), image history, favorites, search, sorting, and metadata management within a responsive user interface.

Smart Waste Classification System
Technologies: TensorFlow, Flask, OpenCV, Python
- Developed a deep learning application to classify waste into six categories using Convolutional Neural Networks (CNNs).
- Implemented image upload, real-time camera detection, and recycling recommendations through a responsive Flask web application.

Apple Disease Detection
Technologies: TensorFlow, Keras, Flask, Python, NumPy, Pillow
- Built a CNN-based application to detect apple leaf diseases from images with high prediction accuracy.
- Designed a Flask dashboard displaying prediction results, confidence scores, and disease information.

AI-Powered People Counting System
Technologies: YOLOv8, OpenCV, Flask, Python
- Developed a real-time people counting system using YOLOv8 object detection and multi-object tracking.
- Built a web dashboard to visualize occupancy statistics and crowd analytics in real time.

Credit Card Fraud Detection System
Technologies: Python, Scikit-learn, XGBoost, Flask, Pandas
- Developed a machine learning system to detect fraudulent credit card transactions using multiple classification algorithms.
- Improved model performance through data preprocessing, feature engineering, and class imbalance handling techniques.

AI Typing Speed Tester
Technologies : Python , Flask , Supabase , PostgreSQL , JavaScript , Chart.js , HTML , CSS
- Developed a full-stack typing speed platform with secure authentication, real-time WPM tracking, analytics, leaderboards, achievements, daily & weekly challenges, user profiles, and XP-based gamification using Flask and Supabase.

SKILLS

Programming Languages
- Python
- SQL
- JavaScript
- HTML
- CSS

Soft Skills
- Problem Solving
- Analytical Thinking
- Team Collaboration
- Communication
- Continuous Learning

Machine Learning
- Machine Learning
- Deep Learning
- Supervised Learning
- Feature Engineering
- Data Preprocessing
- Model Training
- Model Evaluation
- Predictive Modeling

Database & Backend
- Supabase
- SQL
- REST APIs

Artificial Intelligence
- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- Natural Language Processing (NLP)
- Speech Processing
- Computer Vision
- Object Detection
- Image Classification
- Prompt Engineering
- AI Chatbot Development

Libraries & FrameWorks
- Flask
- Streamlit
- Scikit-learn
- TensorFlow
- Keras
- Pandas
- NumPy
- OpenCV

Tools & Platform
- Git
- GitHub
- VS Code
- Jupyter Notebook
- Google Colab
- Kaggle

CERTIFICATES

Artificial Intelligence & Machine Learning with Python — NAVTTC

ACHIEVEMENTS

- Developed AI, Machine Learning, Computer Vision, and Full-Stack Python applications.
- Built real-time AI solutions including fraud detection, object detection, AI assistants, and image generation systems.
- Integrated modern AI technologies such as Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), YOLOv8, and Hugging Face APIs into practical applications.
- Continuously expanding expertise in Artificial Intelligence, Machine Learning, and Generative AI through hands-on projects.